Polynomial-depth mistake converses for neural priors with controlled internal amplification

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Research draft, 22 October 2026

Abstract

Does making few online prediction mistakes force a Bayesian neural-network learner's target to have a small implementation, with polynomial rather than exponential dependence on depth? We prove such a converse for a Gaussian tanh prior restricted by bounded output magnitude and a label-independent certificate controlling propagation of internal noise. The certificate allows compensated layer expansion and includes networks whose internal matrices have operator norm at most one. It is preserved by cloning and direct sums. An exact singleton correction then increases the square root of minimum energy by only a polynomial-depth additive cost. A deterministic robustness compressor turns the resulting quadratic energy bound into an explicit circuit-width bound with fourth-power mistake dependence and no exponent depending on depth. The theorem concerns the limit taking prior variance to zero at each fixed width and then letting width tend to infinity. It handles tied minimizers, posterior sampling, majority prediction, and posterior-confidence failures, in any complete input-reveal order. The original fixed-width converse, and cheap implementations of every small deep Boolean circuit under this stronger restriction, remain open. The main gain-restricted converse and its stated prediction-rule corollaries have Lean proofs.

1 Question and scope

An online learner receives an input, predicts its Boolean label, and then receives the true label. Every label is revealed, including after correct predictions. A converse asks whether a small total mistake count implies that the entire target has a small circuit. It requires sufficient input coverage: observations on a small easy subset cannot constrain the target elsewhere.

For minimum-size Boolean-circuit induction, an incorrect prediction can be repaired by adding a short exception circuit. This immediately bounds circuit size by the mistake count. The analogous neural argument has two difficulties. A large wrong real-valued output may require an expensive correction, and a low-energy neural network need not have a known polynomial-depth compressor. We address these separately: a fixed output cap makes corrections inexpensive, while a stable internal-noise certificate supplies a polynomial-depth compressor.

Our conclusion concerns bounded-weight tanh-network width at the stated depth, rather than unrestricted-depth NAND gate count. The new argument combines an exact singleton construction with the deterministic compressor underlying the earlier polynomial-depth learning result [1]. We reproduce the required compression argument below. The prior restriction is specified in weights and outputs, without reference to training labels, target complexity, or mistake count.

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The order of limits is substantive. We take the strong-prior limit at each fixed width first, and only then let width tend to infinity. This supplies the extra neurons used by an exception branch. We do not assert that the converse holds at every fixed realizable width, or that this limit is one of the finite-exponent width-first posteriors.

2 A prior with controlled internal amplification

Fix $d \geq 1$, $L \geq 2$, $H = L - 1$, $U > 1$, and $\Lambda > 1$. Let $\mathcal{X} = \{-1, 1\}^d$ and $\bar{x} = (x, 1)$. A network with H hidden layers of common width N has

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}) \quad (2 \leq s \leq H), \qquad z_w = v^\top h^H.$$

Its normalized Gaussian energy and readout energy are

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N \|v\|_2^2, \qquad E_v(w) = N \|v\|_2^2.$$

The independent raw Gaussian coordinates have variances $N\sigma^2/d$ in the first layer, σ^2 internally, and σ^2/N in the readout. The density is proportional to $\exp[-\kappa_N/(2\sigma^2)]$.

For diagonal matrices $D_2, \ldots, D_H$ with entries in $[0, 1]$, put

$$B_j(w, D) = v^\top D_H W^H \cdots D_{j+1} W^{j+1} \quad (1 \leq j < H), \qquad B_H(w, D) = v^\top.$$

Define the certified squared hidden-noise gain

$$G_N(w)^2 = N \max_{D_2, \ldots, D_H} \sum_{j=1}^{H} \|B_j(w, D)\|_2^2. \tag{1}$$

The maximum is over a compact finite-dimensional cube, so it exists and is continuous in the weights. For $H = 1$, it is simply E_v.

Let $\mathcal{S}_{N,\Lambda,U}$ be the following open parameter set:

$$\frac{1}{2} < |z_w(x)| < U \quad (x \in \mathcal{X}), \qquad G_N(w)^2 < \Lambda^2 H E_v(w). \tag{2}$$

The prior is the raw Gaussian conditioned on this set. Training further conditions on exact agreement of output signs with the revealed labels. Thus we retain the original global lower-margin convention and add an upper output cap and an internal-amplification certificate.

Remark 2.1 (Interpretation). The diagonals in (1) range over all possible slopes of tanh, not only slopes at the clean forward pass. The condition therefore certifies finite internal disturbances rather than only an infinitesimal Jacobian at one state. Requiring $\|W^s\|_{\mathrm{op}} \leq 1$ at every internal layer is sufficient, since then $G_N^2 \leq H E_v$. The certificate is more flexible: expansion in a layer is allowed when subsequent contractions or readout directions control its effect. It is stronger than checking only the perturbations encountered on a particular data set.

Lemma 2.2 (Certified finite-noise stability)**.** *Add disturbances after every hidden activation and at the readout, and measure them by*

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \|\eta^s\|_2^2 / N + |\eta^{\mathrm{out}}|^2.$$

Every w satisfying the gain condition in (2) obeys

$$|z_w(x;\eta) - z_w(x)| \le \Lambda\sqrt{1 + HE_v(w)}\,\|\eta\|_* .$$

In particular, with $c = 1/(16\Lambda)$ and $r_c(w) = c/\sqrt{1 + HE_v(w)}$, its clipped output $T(z) = \mathrm{clip}(z, -1/2, 1/2)$ changes by at most $1/16$ whenever $\|\eta\|_* \le r_c(w)$.

Proof. The difference of two tanh values is their input difference times a number in $[0, 1]$. Thus the hidden-state differences satisfy $\Delta h^1 = \eta^1$ and $\Delta h^s = D_s W^s \Delta h^{s-1} + \eta^s$ for suitable diagonals D_s. Expanding this recursion gives $\Delta z = \sum_j B_j(w, D)\eta^j + \eta^{\mathrm{out}}$. Cauchy–Schwarz bounds its magnitude by $\sqrt{1 + G_N^2}\,\|\eta\|_*$, and $1 + G_N^2 \le \Lambda^2(1 + HE_v)$ since $\Lambda > 1$. Clipping is one-Lipschitz. □

3 Statement of the converse

For a Boolean function g, let $a_N(g)$ be the infimum of $\kappa_N(w)$ over $w \in \mathcal{S}_{N,\Lambda,U}$ with sign function g, with value $+\infty$ when there is no such implementation. Set $a_\infty(g) = \inf_N a_N(g)$. A minimum-action predictive rule is one supported, at each history, on functions minimizing a_∞ among those agreeing with that history. It may sample such a function, choose one deterministically, or predict by the majority of such a distribution. Confidence failure means that the correct label has probability at most a fixed $\theta \in (0, 1)$.

Let $C_L(f)$ be the smallest integer C such that a depth-L, width-C tanh network has $f(x)z(x) > 1/2$ everywhere and every weight has absolute value at most C. The network witnessing this conclusion need not satisfy the additional prior restriction.

Define explicit budgets

$$\begin{aligned} A_d &= \tfrac{1}{2}\log(4(d+1)), & F_d &= 1 + A_d^2 d(d+1), \\ D_{d,H} &= dF_d + H, & Q_H &= 4(H+65), \\ P_{d,H,U} &= D_{d,H} + Q_H(U+1)^2. \end{aligned}$$

Theorem 3.1 (Polynomial-depth mistake converse)**.** *Let $f : \mathcal{X} \to \{-1, 1\}$ and reveal every point of $\mathcal{X}$, in any order. For any minimum-action predictive rule as above, let m be either its actual total mistake count on a complete-revelation path or its expected total mistake count, using any one of the stated prediction conventions. Put*

$$K = 1 + P_{d,H,U}(m+1)^2.$$

Then

$$C_L(f) \le \left\lceil 65536\Lambda^2 K(1 + HK)[d + \log(400L)] \right\rceil . \tag{3}$$

This is a polynomial bound jointly in d, L, U, Λ, m, with logarithmic factors displayed and with leading fourth-power dependence on m. No exponent or implicit constant depends on depth.

Every subsequential predictive limit obtained by first taking $\sigma \downarrow 0$ at fixed N for the prior (2), and then taking $N \to \infty$, is a minimum-action rule and satisfies the theorem. The same conclusion holds on an infinite iid stream with full support, since every domain point is eventually revealed and repeated points cause no further mistakes.

The theorem also holds if the gain certificate is replaced throughout by $\|W^s\|_{\mathrm{op}} < 1$ for every internal layer. This simpler support has its own minimum energies and predictive limits; its closure

and singleton constructions below are unchanged. Its networks satisfy Lemma 2.2 with any fixed $\Lambda > 1$, for example $\Lambda = 2$.

The statement is a converse. It does not establish that every small Boolean circuit has a low-energy implementation in this support. A fixed output cap controls a true amplitude obstruction in the repair argument; it is not inferred from small sign error or bounded hidden activations. A hard-conditioning prior need not describe ordinary SGD.

4 Cloning, direct sums, and the Gaussian limit

Lemma 4.1 (Closure under cloning and direct sums)**.** *Uniformly clone each neuron of a width-n network k times, divide its outgoing weights by k, and pad with zero neurons to any width $N \geq kn$. Its function is unchanged. If $\lambda = N/(kn)$, then*

$$E'_v = \lambda E_v, \qquad G_N(w')^2 = \lambda G_n(w)^2.$$

The first-layer energy is divided by λ, internal energy is unchanged, and readout energy is multiplied by λ. Thus gain support and output caps are preserved.

For a block-diagonal direct sum of widths n_1, n_2, with outputs added and ambient width $N = n_1 + n_2$,

$$E_v = \sum_{i=1}^{2} \frac{N}{n_i} E_{v_i}, \qquad G_N^2 = \sum_{i=1}^{2} \frac{N}{n_i} G_{n_i}^2.$$

Consequently the strict gain certificate is preserved. The output cap is preserved whenever the resulting real output satisfies it.

Proof. On active clone coordinates the internal matrix is $W^s \otimes (J_k/k)$ and the readout is $v \otimes \mathbf{1}_k/k$. Every path row B_j is the old path row, with diagonal entries replaced by their within-clone averages, tensored with $\mathbf{1}_k^\top/k$. Its squared norm is divided by k. All averaged diagonals in $[0, 1]$ are attainable by taking identical slopes in each clone group, so the supremum is exactly the old one divided by k. Padding has no effect on these path rows. This proves the gain identity; direct calculation gives the remaining cloning energies.

For a direct sum, every path row splits into the two blocks. The slope choices in the blocks are independent, so the supremum of the sum of squared norms is the sum of the two suprema. This proves the identities. For the operator-norm support, cloning preserves internal operator norms and a block-diagonal matrix has norm equal to the maximum of its blocks. □

For an exactly realizable real function u, let $E(u)$ be the infimum energy over all widths satisfying the gain certificate, without a lower margin or upper output constraint. Zero functions may be assigned energy zero by a limit. Lemma 4.1 gives

$$\sqrt{E(u+v)} \leq \sqrt{E(u)} + \sqrt{E(v)}. \tag{4}$$

Indeed, clone two fixed branches into asymptotic proportions $1-\alpha$ and α. Their first-layer energies are multiplied by these proportions, internal energies do not change, and readout energies are divided by them. The total is at most $E(u)/(1-\alpha) + E(v)/\alpha$, after taking infima and approximation limits. Optimizing α proves (4). The same argument works for the operator-norm support.

Proposition 4.2 (Connection with the stated Gaussian learner)**.** *Every Boolean function has finite a_∞, and $a_N(g) \to a_\infty(g)$ as $N \to \infty$. The open-support Gaussian prior's fixed-width strong-prior posterior is supported in the limit on functions of smallest a_N consistent with the history. Every subsequent width-limit accumulation point is supported on smallest a_∞.*

Proof. Finite realizability follows from the exact singleton construction in the next section: at one hidden layer sum the singleton expansions with coefficients $g(x)$ to obtain output g exactly. At greater depth, combine the first-layer features with sufficiently small coefficients to make the second-layer operator norm less than one, obtaining a scalar hidden state of equal magnitude and sign $g(x)$. Contractive repeated copies preserve that sign and common magnitude; the readout restores output exactly to g. Cloning the feature rows with suitable proportions makes the second-layer norm arbitrarily close to its variation-norm bound. Thus all internal operator norms are strictly below one and the output is exactly ± 1, within the fixed cap. A sufficiently large finite width suffices.

Cloning a fixed width-n realization with $k = \lfloor N/n \rfloor$ shows that $\limsup_N a_N(g) \le a_n(g)$, because $N/(kn) \to 1$. Taking the infimum over n and using the reverse inequality from the infimum definition proves convergence.

At fixed width, the event for any given function is open in parameter space. In weighted coordinates the raw Gaussian is $N(0, \sigma^2 I)$. The elementary Gaussian rate calculation yields $2\sigma^2 \log \mu_{N,\sigma}(U_g) \to -a_N(g)$: integrate over a small ball for the lower bound, and use the Gaussian norm tail for the upper bound. The support-normalization factor cancels in ratios. There are only finitely many functions, so strictly higher-energy functions disappear after conditioning on a noiseless history. The same finite-set argument and convergence of a_N prove the final claim. Simultaneous subsequences for all finite distinct-input histories exist because their number is finite. □

5 Exact singleton corrections with controlled amplification

Lemma 5.1 (An inexpensive exact point indicator)**.** *For every $a \in \mathcal{X}$ and $t \in \mathbb{R}$, the function $tI_a(x) = t\mathbf{1}\{x = a\}$ has implementations at depth L with internal operator norms strictly below one and with infimum energy at most*

$$E(tI_a) \le D_{d,H} + Q_H t^2.$$

Proof. There are $d+1$ affine-tanh features $h_0, \dots, h_d$ satisfying

$$I_a = \sum_{j=0}^{d} b_j h_j, \qquad \sum_j |b_j| \le 4, \qquad \|\text{first-layer row } j\|_2^2 \le F_d. \tag{5}$$

To see this, take $h_0 = \tanh(1)$ and

$$h_j(x) = \tanh\left(A_d[a \cdot x - (d - 2j + 1)]\right), \quad 1 \le j \le d.$$

On the shells $s = (d - a \cdot x)/2 \in \{0, \dots, d\}$, the columns $1, (1+h_1)/2, \dots, (1+h_d)/2$ are entrywise within e^{-2A_d} of the staircase columns $1, \mathbf{1}\{s < 1\}, \dots, \mathbf{1}\{s < d\}$. The staircase inverse takes differences and a final endpoint, so its induced ℓ^1 norm is at most two. The perturbation in inverse coordinates has norm at most $2(d+1)e^{-2A_d} = 1/2$. The target singleton has staircase coefficient norm one; a Neumann-series inverse gives coefficients of total magnitude at most two. Expanding the affine columns and using $1 = h_0/\tanh(1)$ increases that norm by a factor less than two. The row bound follows from d coefficients of magnitude A_d and a bias of magnitude at most $A_d d$. This proves (5).

Write $V = \sum_j |b_j| \le 4$ and omit zero coefficients. At large width clone feature j in proportion $p_j = |b_j|/V$. If $H = 1$, split its readout coefficient tb_j equally among its copies. The limiting readout energy is $t^2 \sum_j b_j^2/p_j = t^2 V^2 \le 16t^2$, and first-layer energy is at most dF_d.

If $H \ge 2$, let each second-layer neuron receive $b_j/8$ times feature j, distributed across its copies. The resulting state is $a_2 I_a$ with $a_2 = \tanh(1/8)$. The second-layer squared operator norm and

Frobenius energy both tend to $V^2/64 \le 1/4$, so they are below one at all sufficiently accurate finite allocations. Later hidden matrices are constant matrices $\beta J/N$, where $\beta = 1 - 1/(2H) < 1$. Thus every internal matrix is strictly contractive, and the state is $a_s I_a$, where $a_s = \tanh(\beta a_{s-1})$. Readout entries are $t/(Na_H)$.

For $y \ge 0$, $\tanh y \ge y/\sqrt{1+y^2}$, because $\operatorname{arsinh} y \le y$ and $\tanh(\operatorname{arsinh} y) = y/\sqrt{1+y^2}$. Consequently

$$a_2^{-2} \le 65, \qquad a_s^{-2} \le 1 + \beta^{-2} a_{s-1}^{-2}, \qquad a_H^{-2} \le \beta^{-2H}(65+H) \le 4(65+H) = Q_H.$$

The last inequality uses $-\log(1 - 1/(2H)) \le 1/(2H-1)$ and $H \ge 2$. The first-layer cost is at most dF_d; the total internal cost is at most H in the limiting construction; readout energy is at most $Q_H t^2$. Every finite allocation computes the singleton exactly, so no approximation consumes any pre-existing margin slack. □

A constant output $+1$ or -1 also has a strictly contractive implementation of energy at most $d + 5H + 8$. Use identical first-layer features $\tanh(1)$, the same contractive relays, and a readout restoring unit magnitude. The inverse-square calculation above bounds the required readout energy by $4(H+2)$. In particular, the initial minimum action is at most $P_{d,H,U}$.

6 Mistakes bound the minimum action

Let K_t be the smallest a_∞ among functions consistent with the first t revealed labels. This minimum exists, since the domain has finitely many functions and all have finite action. It is nondecreasing. If $K_t > K_{t-1}$, every function minimizing the previous action has the wrong label at the new input. Such a round is a forced mistake for every prediction convention in Theorem 3.1. This remains true with arbitrary mixtures of tied minimizers.

Choose a minimizing function before such an increase and a capped real-output realization z with energy arbitrarily close to its action. If the next input is a with label y, replace its output by

$$z_{\text{new}}(x) = z(x) + (y - z(a)) I_a(x).$$

Every other output is unchanged, and the new value at a is exactly $y \in \{-1, 1\}$. All previous labels, the strict lower margin, and the upper cap remain valid. Since $|z(a)| < U$, the singleton amplitude is at most $U + 1$. Lemmas 4.1 and 5.1, followed by approximation of the energy infima, give

$$\sqrt{K_t} \le \sqrt{K_{t-1}} + \sqrt{P_{d,H,U}}$$

at an increase step. At other steps $K_t = K_{t-1}$. If B is the number of increase steps after full revelation, then

$$a_\infty(f) \le P_{d,H,U}(B+1)^2 \le P_{d,H,U}(M+1)^2, \tag{6}$$

where M is total mistakes under any stated convention.

For random orders or random predictions, the same conclusion holds with $m = \mathbb{E}M$. Invert (6) first: $M \ge \sqrt{a_\infty(f)/P_{d,H,U}} - 1$ on every complete path. The right side is deterministic, so expectation and rearrangement give $a_\infty(f) \le P_{d,H,U}(m+1)^2$. This does not reverse Jensen's inequality. Predictable adaptive input orders are allowed, provided all points are eventually revealed.

7 Polynomial-depth reconstruction from energy

For completeness we give the deterministic compression statement needed to turn (6) into circuit complexity. It is the pruning-and-Bernstein construction of [1], used without its Gaussian or statistical parts.

Lemma 7.1 (Robust energy compression)**.** *Suppose a width-N, depth-L tanh network has $\kappa_N(w) \le K$, where $K \ge 1$, and its clipped outputs change by at most $1/16$ under all disturbances of norm at most r, where $0 < r \le 1/6$. Put*

$$m_* = \left\lceil 256 K r^{-2} [d + \log(400L)] \right\rceil .$$

There is an actual width-m_*, depth-L tanh network with energy at most $4K$, weights bounded in magnitude by $2\sqrt{m_* K}$, and clipped outputs differing from the original's by at most $1/16$ on the entire Boolean cube.

Proof. Write $k_s = \|W^s\|_F^2$ for internal layers and $k_L = N\|v\|_2^2$, so their sum is at most K. In each internal row and the readout, delete entries of magnitude above $8\sqrt{K}\,\|\mathrm{row}\|_2/(r\sqrt{N})$; zero rows need no change. The absolute removed row mass is at most its squared norm divided by this threshold. On a vector in $[-1,1]^N$, pruning therefore has normalized stage error at most $r\sqrt{k_s/K}/8$.

At each hidden layer independently sample m_* uniform source indices with replacement. Keep sampled first-layer rows, and multiply subsequent sampled source weights by N/m_*. Work first with a virtual pass retaining all destination rows. Its clean virtual hidden states remain in $[-1,1]^N$ and are independent of the fresh source indices at the next stage. For one uniform source index I, the internal vector summand is $Y = N\overline{W}^s_{:,I}\widetilde{h}_I$. It obeys

$$\mathbb{E}\,\|Y - \mathbb{E}Y\|_2^2 \le N k_s, \qquad \|Y - \mathbb{E}Y\|_2 \le 16\sqrt{NKk_s}/r.$$

We use the Hilbert-space Bernstein inequality [3, Appendix A, Corollary A.2]: independent centered Hilbert-space variables X_i with $\|X_i\| \le b$ and $\sum_i \mathbb{E}\,\|X_i\|^2 \le V$ satisfy

$$\Pr\left\{ \left\| \sum_i X_i \right\| \ge t \right\} \le 2\exp\left[-\frac{t^2}{2(V + bt/3)} \right].$$

For the sample average at tolerance $r\sqrt{Nk_s/K}/8$, this bounds the sampling failure probability by $2\exp[-m_* r^2/(256K)]$. The readout calculation is the same without factors of $\sqrt{N}$. A union bound over at most L stages and 2^d inputs is less than $1/200$ by the chosen budget.

On success the pruning and sampling errors combine to give normalized stage errors at most $r\sqrt{k_s/K}/4$. Define postactivation disturbances as the difference between each virtual state and tanh applied to the original weights on the preceding virtual state. One-Lipschitzness makes the total disturbance norm at most $r/4$, uniformly on the cube. A readout disturbance makes the virtual output exact. Thus its clipped output differs from the original by at most $1/16$.

The stored sampled coordinates form an ordinary width-m_* network. Independent index lists in adjacent layers show that its expected normalized energy is at most K: the first-layer contribution is unchanged in expectation, and every remaining contribution is at most its original value. Hence its energy is at most $4K$ with probability at least $3/4$. The energy and all-input-noise events intersect. Choose one such network. The energy bound gives every coordinate magnitude at most $2\sqrt{m_* K}$, including the first layer since $d \ge 1$. □

To finish Theorem 3.1, choose an admissible realization of f with energy below $K = 1 + P_{d,H,U}(m+1)^2$. Lemma 2.2 applies at radius

$$r = \frac{1}{16\Lambda\sqrt{1+HK}},$$

which is at most $1/6$ and no greater than the network's own certified radius. Lemma 7.1 gives precisely the width in (3). Since the original output has signed margin above $1/2$, its signed clipped output is exactly $1/2$. The compressed output has signed margin at least $7/16$. Multiply its readout by two to obtain margin at least $7/8$, strictly above $1/2$. All weights are then at most $4\sqrt{m_* K} \le m_*$, because the displayed budget has $m_* \ge 16K$. This proves $C_L(f) \le m_*$.

For fixed U, Λ, the order of the bound is, explicitly,

$$O\Big(\Lambda^2 L[d^3 \log^2(2+d) + L(U+1)^2]^2 (m+1)^4 [d + \log(2L)]\Big).$$

The exact ceiling in (3) includes the harmless lower-order terms. In particular the degree does not grow with L.

8 Interpretation and remaining questions

The prior conditions are implementational assumptions. A function may have both supported and unsupported realizations. Every Boolean function has some supported realization, but the construction proving this can be exponentially expensive in input dimension. We have not proved that every width-C, depth-L Boolean circuit has a supported implementation with energy polynomial in C, d, L. Thus this paper supplies a polynomial-depth *converse*; it does not claim a two-sided polynomial-depth circuit prior for all small Boolean circuits.

The global gain certificate is sufficient, not necessary for stable predictions. Saturated tanh circuits can be stable near their actual states while failing a certificate that allows every diagonal slope in $[0, 1]$. The simpler layerwise nonexpansiveness restriction is stronger still. Understanding useful weaker restrictions with the same closure properties is a natural next question.

A uniform output cap is likewise substantive. Without it the correction amplitude can scale like the square root of current energy, turning the additive recurrence for its square root into a multiplicative one. The present proof does not remove that obstruction. Finally, direct sums need spare neurons. A fixed-width converse needs a way to create that space without repeatedly paying a constant energy factor, while retaining the prior's support restriction. That step is not supplied here.

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